Title:
From Particle Collisions to Navier–Stokes: Accuracy and Performance of a Lattice Boltzmann Solver

Describing the project:
This project tests how accurately a lattice Boltzmann fluid simulator reproduces known flows, and how much accuracy simple wall boundaries cost at practical grid resolutions.

Abstract:
Computational fluid dynamics usually starts from the Navier–Stokes equations, nonlinear partial differential equations that are difficult to solve directly. The lattice Boltzmann method (LBM) instead evolves discretized particle velocity distributions on a grid, relaxing them toward a discretized Maxwell–Boltzmann equilibrium. Through the Chapman–Enskog expansion, these local collide-and-stream rules recover Navier–Stokes behavior at low Mach number, with viscosity set by a single relaxation time. This project implements a D2Q9 LBM solver with the Bhatnagar–Gross–Krook (BGK) collision operator in C++ and an interactive OpenGL visualization. It asks two questions: how faithfully does this kinetic model reproduce known fluid behavior, and how much accuracy is lost to the simple staircase bounce-back treatment of curved walls? Validation proceeds in three stages. First, channel flow is compared with the analytical Poiseuille profile to test the relationship between relaxation time and viscosity, including the relaxation-time dependence of the effective wall location. Second, flow past a circular cylinder in a low-blockage domain is simulated at Reynolds numbers from 50 to 150, and the vortex-shedding frequency, extracted by Fourier analysis, is compared with experimental Strouhal numbers. Third, drag coefficients computed by momentum exchange are compared with the confined-channel Schäfer–Turek benchmark. Because fine grids are computationally expensive, the solver is being ported to the GPU, with throughput and effective memory bandwidth measured against the CPU implementation. This enables a grid convergence study quantifying how staircase boundary error scales with resolution. Preliminary results will be presented alongside the remaining validation work.

Sources:

Krüger, T., Kusumaatmaja, H., Kuzmin, A., Shardt, O., Silva, G., & Viggen, E. M. (2017). The lattice Boltzmann method: Principles and practice. Springer.
Qian, Y. H., d'Humières, D., & Lallemand, P. (1992). Lattice BGK models for Navier–Stokes equation. Europhysics Letters, 17(6), 479–484.
Schäfer, M., & Turek, S. (1996). Benchmark computations of laminar flow around a cylinder. In E. H. Hirschel (Ed.), Flow simulation with high-performance computers II (pp. 547–566). Vieweg+Teubner.
Williamson, C. H. K. (1996). Vortex dynamics in the cylinder wake. Annual Review of Fluid Mechanics, 28, 477–539.
He, X., Zou, Q., Luo, L.-S., & Dembo, M. (1997). Analytic solutions of simple flows and analysis of nonslip boundary conditions for the lattice Boltzmann BGK model. Journal of Statistical Physics, 87, 115–136.

Vomit:

Chen, S., & Doolen, G. D. (1998). Lattice Boltzmann method for fluid flows. *Annual Review of Fluid Mechanics, 30*, 329–364.

Krüger, T., Kusumaatmaja, H., Kuzmin, A., Shardt, O., Silva, G., & Viggen, E. M. (2017). *The lattice Boltzmann method: Principles and practice*. Springer.

Qian, Y. H., d'Humières, D., & Lallemand, P. (1992). Lattice BGK models for Navier–Stokes equation. *Europhysics Letters, 17*(6), 479–484.

He, X., Zou, Q., Luo, L.-S., & Dembo, M. (1997). Analytic solutions of simple flows and analysis of nonslip boundary conditions for the lattice Boltzmann BGK model. *Journal of Statistical Physics, 87*, 115–136.

Schäfer, M., & Turek, S. (1996). Benchmark computations of laminar flow around a cylinder. In *Flow simulation with high-performance computers II* (pp. 547–566). Vieweg+Teubner.
Williamson, C. H. K. (1996). Vortex dynamics in the cylinder wake. *Annual Review of Fluid Mechanics, 28*, 477–539.

Williamson, C. H. K. (1996). Vortex dynamics in the cylinder wake. *Annual Review of Fluid Mechanics, 28*, 477–539.

Old:

Computational fluid dynamics usually begins with the Navier–Stokes equations, a set of nonlinear partial differential equations that are difficult to solve directly. The lattice Boltzmann method (LBM) takes a different approach. Instead of treating the fluid as a continuum, it evolves discretized particle velocity distributions on a grid, relaxing them toward a discretized Maxwell–Boltzmann equilibrium. Through the Chapman–Enskog expansion, these local collide-and-stream rules recover Navier–Stokes behavior at low Mach number, with viscosity set by a single relaxation time. This project implements a D2Q9 lattice Boltzmann solver with the

Bhatnagar–Gross–Krook (BGK) collision operator in C++, with interactive OpenGL visualization. It asks two questions: how faithfully does this kinetic model reproduce known fluid behavior, and how much accuracy is lost to the simple staircase bounce-back treatment of curved walls? Validation proceeds in three stages. First, pressure-driven channel flow is compared with the analytical Poiseuille profile to test the predicted relationship between relaxation time and viscosity, including the known dependence of the effective wall location on relaxation time. Second, flow past a circular cylinder in a wide, low-blockage domain is simulated for Reynolds numbers from about 50 to 150, and the vortex-shedding frequency, extracted by Fourier analysis, is compared with experimental Strouhal number measurements. Third, drag coefficients computed by the momentum-exchange method are compared with the confined-channel Schäfer–Turek benchmark. Because resolving these flows at fine grid spacing is computationally expensive, the solver is also being ported to the GPU, and its throughput and effective memory bandwidth are measured against the CPU implementation. This enables a grid convergence study quantifying how staircase boundary errors scale with resolution. Preliminary results will be presented alongside the remaining validation work.